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Graded Quiz: Introduction to Vector Databases and Chroma DB
Graded Assignment • 21 min

Due Aug 22, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 71%. We keep your highest score.

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Skill progress from this assignment Beta

Advanced RAG System Implementation	+15 XP 251/400 XP

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1. **How do vector databases enhance recommendation systems compared to traditional databases?** 1 / 1 point

○ By supporting larger volumes of data than traditional databases
● By enabling efficient similarity searches for complex data types
○ By providing a more secure data storage solution
○ By storing data in a format that simplifies complex queries

✓ Nice work

Vector databases are designed to perform efficient similarity searches, which are crucial for recommendation systems dealing with complex data types.

2. **What is a key difference between vector databases and traditional databases?** 1 / 1 point

● Vector databases are designed to handle embeddings.
○ Vector databases use relational tables for data storage.
○ Traditional databases are optimized for numerical data types.
○ Traditional databases are more efficient for similarity searches.

✓ Nice work

Vector databases are specifically designed to store and manage vector embeddings that capture the semantic meaning of the underlying data.

3. **Which of the following statements differentiates between in-memory vector databases and other types of vector databases in terms of storage?** 1 / 1 point

○ In-memory vector databases are a type of database that do not support vector search.
○ In-memory vector databases are slower in query responses compared to other types.
● In-memory vector databases store vectors directly in memory, which can lead to faster query responses.
○ In-memory vector databases store vectors on disk, which enhances data persistence.

✓ Nice work

Storing vectors directly in memory allows quicker access and faster query responses than other storage methods. However, this approach sacrifices persistence, as in-memory databases do not retain data after a system shutdown or crash.

4. **Which metric measures the distance between vectors based on direction but not their magnitude?** 1 / 1 point

○ Inner product distance
○ Dot product distance
● Cosine distance
○ Euclidean distance

✓ Nice work

Cosine distance depends on the angle between vectors, not their magnitudes.

5. **What does the direction of a vector embedding typically represent?** 1 / 1 point

○ The strength of the item being represented
○ The distance between two unrelated items
● The topic or semantic meaning of the item
○ The frequency of the item in the dataset

✓ Nice work

The direction of a vector embedding typically captures the core meaning or topic of the item it represents.

6. **Which of the following statements differentiates between the roles of indexing and querying in Chroma DB architecture?** 1 / 1 point

○ Indexing is responsible for data input, whereas querying is responsible for data output.
○ Indexing is used to encrypt data, whereas querying is used to decrypt it.
○ Indexing and querying both serve the same purpose of data retrieval in Chroma DB.
● Indexing organizes data for efficient retrieval, whereas querying retrieves data based on semantic similarity.

✓ Nice work

Indexing in Chroma DB is vital for organizing data to make retrieval efficient, whereas querying involves retrieving data based on the query and the preselected distance metric.

7. **Which of the following statements most accurately defines the role of vector embeddings in Chroma DB's similarity search process?** 1 / 1 point

○ Vector embeddings are used to store metadata about the data entries.
○ Vector embeddings are used to encrypt data for security purposes in Chroma DB.
○ Vector embeddings are primarily used to increase the speed of database queries.
● Vector embeddings are used to convert data into a format that enables similarity comparison.

✓ Nice work

Vector embeddings are crucial in Chroma DB. They transform data into a numerical format that allows for effective similarity searching.